Analyst : Benjamin Liu

Date : 4/5/2026

- **Zeek : A** powerful **framework for network traffic analysis and security monitoring. [**https://github.com/zeek/zeek**]**
- **RITA** (Real Intelligence Threat Analytics): an open source framework for network traffic analysis.
- **FQDN (Fully Qualified Domain Name)**

Q1) Looking at the RITA HTML Report (Log_2V), what is the IP of the attacker's C2 server? Provide the number of connections as well. (Format: IP Address, Connections) *(3 points)*

138.197.117.74, 8440 Correct! ✔

In the RITA HTML_Report(LOg_2V), we notice that the connections are far more than others.

Score	Source	Destination	Connections	Avg. Bytes	Intvl. Range	Size Range	Intvl. Mode	Size Mode	Intvl. Mode Count	Size Mode Count	Intvl. Skew	Size Skew	Intvl. Dispersion	Size Dispersion	Total Bytes
0.996	10.234.234.100	138.197.117.74	8440	1317.000	8	935	10	544	7306	8312	0.000	0.000	0	0	11115975

Q2) What is the cloud infrastructure being used for the C2 server? (Format: Infrastructure) *(3 points)*

DigitalOcean Correct! ✔

We have known the C2 server IP. Then We search the ip at Ipinfo.com.

138.197.117.74 Star

Clifton, New Jersey, US | Cloud | Hosting | Webserver

- Summary
- Geolocation
- Privacy
- ASN
- Company
- Abuse

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Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	138.197.112.0/20
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

Q3) Looking at the RITA User Agent report, what is the system that corresponds to the connection count in Q1? (Format: User Agent String) *(3 points)*

Mozilla/5.0 (compatible; MSIE 9.0; Windows Correct! ✔

Empty user agent string	4000
Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; Trident/5.0)	8439

Q4) Let's look at Log_1D, what is the low-profile subdomain with the absurd amount of requests? Provide the number of requests as well. (Format: Subdomain, Request) *(3 points)*

cat.nanobotninjas.com, 82920

Correct! ✔

file:///home/ubuntu/Desktop/Investigation/RITA HTML Report/Log_1D/dns.html

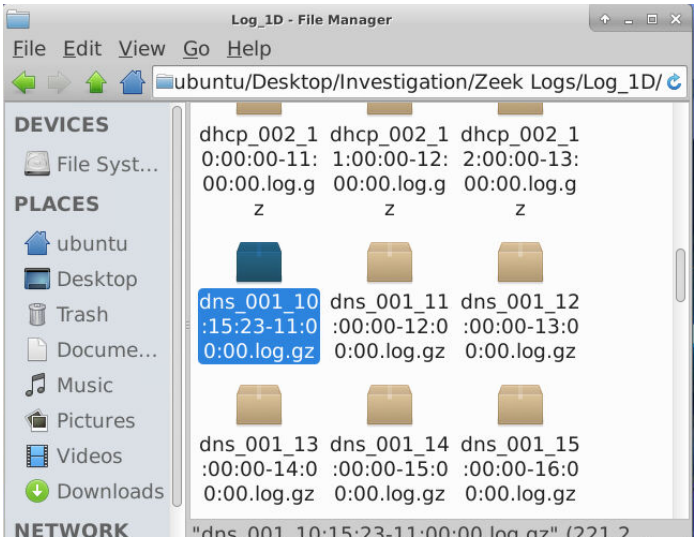
RITA | Viewing: Log_1D | Beacons | Beacons FQDN | Strobes | DNS | BL Source IPs | BL Dest. IPs | BL

Subdomain Count	Visited	Domain
30189	82920	nanobotninjas.com
30189	82920	cat.nanobotninjas.com

Q5) In the Zeek logs (Log_1D), we can see a large number of DNS TXT record requests for a private IP for the subdomain found in Q4. List the IP. (Format: IP Address) *(3 points)*

10.234.234.105

Submit



```
Terminal
File  Edit  View  Search  Terminal  Help
#separator \x09
#set_separator  ,
#empty_field    (empty)
#unset_field    -
#path   dns
#open   2017-03-20-10-15-23
#fields ts      uid     id.orig_h       id.orig_p       id.resp_h       id.resp_p       proto   trans_id        query   qclass  qclass_name     qtype   qtype_name      rcode   rcode_name      AA      TC      RD
      RA      Z       answers TTLs    rejected
#types  time    string  addr    port    addr    port    enum    count   string  count   string  count   string  bool    bool    bool    bool    count   vector[string]  vector[interval]        boo
l
1490022923.690245       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     18453   03b001050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 acd601050cfbab0d05       59.000000       F
1490022924.862270       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     4269    747001050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 0d7601050cfbab0d05       59.000000       F
1490022926.034271       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     15432   6e3701050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 787a01050cfbab0d05       59.000000       F
1490022927.206294       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     5989    65ba01050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 ec4a01050cfbab0d05       59.000000       F
1490022928.378265       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     2528    7d9501050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 1cb301050cfbab0d05       59.000000       F
1490022929.550197       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     19717   571b01050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 0b3001050cfbab0d05       59.000000       F
1490022930.722256       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     11032   76a601050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 693d01050cfbab0d05       59.000000       F
1490022931.894193       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     23627   3f7001050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 bcf201050cfbab0d05       59.000000       F
1490022933.066245       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     29752   6b4201050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 6e5301050cfbab0d05       59.000000       F
1490022934.238253       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     11354   447301050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 a79b01050cfbab0d05       59.000000       F
1490022935.410223       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     31511   5d1201050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 8f3a01050cfbab0d05       59.000000       F
1490022936.582336       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     9309    142f01050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 a95001050cfbab0d05       59.000000       F
1490022937.754199       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     6110    02a101050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 3fa701050cfbab0d05       59.000000       F
1490022938.926248       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     21101   72b801050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 319b01050cfbab0d05       59.000000       F
1490022940.130246       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     25358   62b601050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 d6d501050cfbab0d05       59.000000       F
1490022941.302189       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     31157   501801050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 ddde01050cfbab0d05       59.000000       F
1490022942.474194       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     21701   39e201050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 214801050cfbab0d05       59.000000       F
1490022943.646189       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     19411   7a8c01050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 977701050cfbab0d05       59.000000       F
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
1 Top
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So we get the IP of

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l
1490022923.690245       CbEtxF16rCk0utMKw2      10.234.234.105  56122   8.8.8.8 53      udp     18453   03b001050c0d05fbab.cat.nanobotninjas.com        1       C_INTERNET      16      TXT     0       NOERROR F
    F       T       T       0       TXT 18 acd601050cfbab0d05       59.000000       F
```

Q6) This is an unusual activity. To avoid cached results on the intermediate DNS server(s), a certain value prepended to the subdomain is being changed. What numeral system is being used for the value that changes? It seems like a Base 16 numbering system. (Format: value) *(5 points)*

hex

Submit

"Base16", "16" are not right answer.

```
03b001050c0d05fbab.cat.nanobotninjas.com
747001050c0d05fbab.cat.nanobotninjas.com
6e3701050c0d05fbab.cat.nanobotninjas.com
65ba01050c0d05fbab.cat.nanobotninjas.com
7d9501050c0d05fbab.cat.nanobotninjas.com
571b01050c0d05fbab.cat.nanobotninjas.com
76a601050c0d05fbab.cat.nanobotninjas.com
3f7001050c0d05fbab.cat.nanobotninjas.com
6b4201050c0d05fbab.cat.nanobotninjas.com
447301050c0d05fbab.cat.nanobotninjas.com
5d1201050c0d05fbab.cat.nanobotninjas.com
142f01050c0d05fbab.cat.nanobotninjas.com
02a101050c0d05fbab.cat.nanobotninjas.com
```

Q7) Judging from the logs (Log_1D), what tool was the attacker using? A tool to help C2s channel over the DNS protocol. (Format: tool) *(5 points)*

Answer: dnscat2

- **dnscat2** is a well-known tool specifically designed to create **C2 channels over DNS**, and its traffic patterns match exactly what you described
  - Uses **high-frequency DNS queries**
  - Encodes data inside **subdomain labels**
  - Uses **A records + TXT records** for command exchange
  - Produces **long, random-looking subdomains**
  - Is widely used in training labs and red-team exercises